

Evidence that stress-induced changes in surface temperature serve a thermoregulatory function

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ABSTRACT

The fact that body temperature can rise or fall following exposure to stressors has been known for nearly two millennia; however, the functional value of this phenomenon remains poorly understood. We tested two competing hypotheses to explain stress-induced changes in temperature, with respect to surface tissues. Under the first hypothesis, changes in surface temperature are a consequence of vasoconstriction that occur to attenuate blood loss in the event of injury and serve no functional purpose per se; defined as the 'haemoprotective hypothesis'. Under the second hypothesis, changes in surface temperature reduce thermoregulatory burdens experienced during activation of a stress response, and thus hold a direct functional value: the 'thermoprotective hypothesis'. To understand whether stress-induced changes in surface temperature have functional consequences, we tested predictions of these two hypotheses by exposing black-capped chickadees ($n=20$) to rotating stressors across an ecologically relevant ambient temperature gradient, while non-invasively monitoring surface temperature (eye region temperature) using infrared thermography. Our results show that individuals exposed to rotating stressors reduce surface temperature and dry heat loss at low ambient temperature and increase surface temperature and dry heat loss at high ambient temperature, when compared with controls. These results support the thermoprotective hypothesis and suggest that changes in surface temperature following stress exposure have functional consequences and are consistent with an adaptation. Such findings emphasize the importance of the thermal environment in shaping physiological responses to stressors in vertebrates, and in doing so, raise questions about their suitability within the context of a changing climate.

KEY WORDS: Thermoregulation, Haemodynamics, Thermography, *Poecile atricapillus*

INTRODUCTION

Changes in body temperature following perception of a stressor were first reported almost 2000 years ago. In the 2nd century CE, for example, Galen described the presence of an 'ephemeral fever' in humans that was thought to be evoked by an abundance of humour (Yeo, 2005); a concept later expanded upon by Ibn Sina (11th century CE) who described 'fevers' driven by emotions, including grief, anger and dread (reviewed in Parviz et al., 2013). To date,

changes in body temperature following exposure to stressors have been reported across numerous non-hominid species (i.e. lizards: Cabanac and Gosselin, 1993; turtles: Cabanac and Bernieri, 2000; birds: Greenacre and Lusby, 2004, and fish: Rey et al., 2015) and the area has attracted significant research attention. Despite such long-standing recognition in philosophical and scientific literature, and near vertebrate-wide conservation (but see Cabanac and Laberge, 1998; Jones et al., 2019), the functional significance of stress-induced changes in body temperature remain elusive. Indeed, while many studies have done well to uncover proximate mechanisms underpinning this phenomenon (i.e. mediation by peripheral vasoconstriction, pyrogenic cytokines, prostaglandins or glucocorticoids; Yokoi, 1966; reviewed in Oka et al., 2001; and Jerem et al., 2018, respectively), remarkably few have empirically interrogated a functional role of changing one's body temperature, per se, in response to stress exposure (Cabanac and Gosselin, 1993).

Thermal responses to stress exposure can be broadly categorized according to their position of occurrence within an organism (i.e. at core tissues or surface tissues). Of these, core temperature responses have, perhaps, received the greatest theoretical attention with respect to functional significance (reviewed in Oka et al., 2001; Oka, 2018). For example, some immunological studies have posited that changes in core body temperature following stress exposure represent a true 'fever' (Singer et al., 1986; Sanches et al., 2002) and endow individuals with a defensive advantage if they experience injury and pathogen exposure during stress exposure (discussed in Oka et al., 2001). Others, however, have contested this hypothesis by failing to substantiate immune mediation of core temperature responses to stress (Long et al., 1990b; Soszynski et al., 1998; Hiramoto et al., 2009; Vinkers et al., 2009). At the level of surface tissues, however, a functional role of stress-induced changes in temperature, per se, appears yet to be raised (though briefly discussed in Herborn et al., 2018). Indeed, dominant theory explaining stress-induced changes in surface temperature posits that this phenomenon is merely a consequence of haemetic redistribution driven by peripheral vasoconstriction (Jerem et al., 2015; Jerem et al., 2018; Nord and Folkow, 2019) and holds no direct functional role; rather, it is haemetic redistribution, but not thermal modulation, that carries functional significance by attenuating blood loss in the event of injury (as long shown following haemorrhage by McGuigan and Atkinson, 1921; Freeman, 1932; Darlington et al., 1986).

Despite an absence of functionally guided research, studies describing stress-induced changes in surface temperature do allude to a functional value of this phenomenon. Nord and Folkow (2019), for example, reported that while the skin temperature of Svalbard rock ptarmigans (*Lagopus muta hyperborea*) typically falls after handling, the magnitude of the skin temperature response varies according to ambient temperature (T_a); specifically, ptarmigans handled at low temperature (-20°C) display a larger change in skin temperature than those handled at warmer temperatures (0°C). An

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